ITA0522 - COMPUTER VISION

CO 2 - ASSIGNMENT – 2

APPLY, COMPARE AND JUSTIFY IMAGE ENHANCEMENTAND SEGMENTATION TECHNIQUES

Course Outcome: CO2 – Apply image enhancement and segmentation techniques for extracting meaningful regions from images.

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ABSTRACT

This report presents a complete design study for a smart document scanner intended to digitize aged college question papers and lecture notes. The source documents may contain yellowed paper, faded ink, stains, ink blots, dust, punch holes, camera grain, scanner stripes and non-uniform illumination. The scanner must enhance text for printing and optical character recognition (OCR), locate the page boundary, identify ruled lines and table structures, and annotate physical defects for staff inspection.

Ten assignment tasks are addressed. Point transformations, histogram equalization, spatial smoothing, sharpening, frequency-domain low-pass and high-pass filtering, point/line/edge detection, thresholding, region segmentation and adaptation to high-resolution phone photographs are discussed and compared. Each task includes mathematical foundations, parameter assumptions, implementation guidance, comparative analysis, likely failure modes and a decision justified against the stated requirements.

Because no image dataset or executed measurement logs accompanied the assignment PDF, all example numerical tables in this report are explicitly marked as illustrative assumed values. They are internally consistent design examples, not claims of execution. A reproducible Python/OpenCV framework and evaluation formulas are included so that actual results can replace the assumed values. The proposed baseline for the original A4 grayscale dataset is conservative CLAHE, light median/gaussian denoising, controlled unsharp masking where required, adaptive thresholding for shadowed pages, Canny/Hough-based border detection and contour-based region annotation. For the changed 12-megapixel phone-photo requirement, the recommended decision is MODIFY: resize for detection and processing, preserve colour information for blue annotations, detect the page quadrilateral, apply perspective correction, and use selective high-resolution refinement.

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
CLAHE	Contrast Limited Adaptive Histogram Equalization
CDF	Cumulative Distribution Function
CER	Character Error Rate
DCT	Discrete Cosine Transform
DFT	Discrete Fourier Transform
DIBCO	Document Image Binarization Contest
DPI	Dots Per Inch
HED	Holistically-Nested Edge Detection
HPF	High-Pass Filter
ILPF	Ideal Low-Pass Filter
LPF	Low-Pass Filter
OCR	Optical Character Recognition
PSNR	Peak Signal-to-Noise Ratio
SSIM	Structural Similarity Index
IoU	Intersection over Union
DC	Direct Current / zero-frequency component
RGB	Red, Green and Blue
HSV	Hue, Saturation and Value
CER	Character Error Rate
WER	Word Error Rate

1. INTRODUCTION AND PROBLEM DEFINITION

1.1 Background

Libraries preserve printed material that may be physically old, faded or damaged. Digitization improves access, reduces repeated handling and supports searchable archives. A mobile photograph or old flatbed scan is not automatically suitable for OCR: shadows, uneven illumination, low contrast, stains, dust and geometric distortion can interfere with character recognition. The proposed smart scanner therefore performs image enhancement and segmentation before OCR and archival storage.

1.2 Problem statement

The system receives 8-bit grayscale A4 pages of 2480 × 3508 pixels at 300 dpi. Text strokes are approximately 2–4 pixels wide. Most image values lie between 90 and 180, while faded ink is only 30–50 grey levels darker than the yellowed paper. Phone photographs may have a bright centre and a shadow on one side. Noise includes camera grain with standard deviation approximately 6 grey levels, about 1% salt-and-pepper dust specks and faint horizontal scanner stripes every six pixel rows.

1.3 Functional requirements

- Enhance faded text while preserving small punctuation and pencil marks.
- Separate foreground text from paper with a target Dice score of at least 0.90.
- Find the page border and estimate four page corners for cropping.
- Detect horizontal, vertical and slightly tilted ruled/table lines.
- Mark dust specks, punch holes, tea stains and ink blots for staff review.
- Process each original page in no more than 3 seconds on the specified lab PC.

1.4 Proposed system architecture

The system is organized as a sequence of stages: image acquisition, grayscale/colour analysis, illumination correction, denoising, contrast enhancement, optional sharpening, geometric page detection, line and edge detection, binarization, region annotation, cropping, OCR and quality reporting. Each stage is modular so that the scanner can skip a stage when its benefit is not demonstrated.

2. SYSTEM REQUIREMENTS AND ASSUMPTIONS

Item	Given requirement	Design interpretation
Image size	2480 × 3508, 8-bit grayscale	Approximately 8.70 megapixels per page
Resolution	300 dpi	Text strokes are small; avoid large smoothing kernels
Text stroke width	2–4 pixels	Use 3×3 or carefully tuned filters
Ink-paper difference	30–50 levels	Moderate local contrast enhancement is appropriate
Camera grain	$\sigma \approx 6$	Denoise gently to avoid destroying thin strokes
Dust specks	About 1% salt-and-pepper	Median filtering and connected-component analysis
Scanner stripes	Every 6 rows	Inspect spectrum and use cautious directional suppression
Target Dice	≥ 0.90	Evaluate against manually verified binary masks
Runtime	≤ 3 seconds/page	Measure on the stated CPU; avoid heavy neural models in baseline
Hardware	4 CPU cores, 16 GB RAM, no GPU	Use OpenCV optimized CPU routines and bounded image copies

2.1 Additional assumptions

- The original grayscale page images are stored as lossless PNG or TIFF during experiments; JPEG is retained only for the phone-photo adaptation study.
- At least 10 local pages are collected, and public DIBCO samples are used where licensing and format permit.

- For numerical examples, the foreground mask is defined as text pixels = 1 and paper = 0.
- Illustrative page labels P01–P10 represent a mixture of light, normal, shadowed and stained pages.
- A clean reference page is selected from the same document family for histogram matching.
- A 2-pixel tolerance is used when matching detected edge pixels to ground truth, as required.
- All reported assumed metrics are examples only; actual timings depend on the CPU, OpenCV build, image I/O and parameter settings.

2.2 Risk and integrity note

The assignment specifically requires measured results from code execution. Since the report is prepared without the actual 10-page dataset and without running experiments on the user's lab PC, numeric results are labelled illustrative. The code appendix provides the measurement functions and logging structure. Replace every assumed table with the output from the real dataset before submission.

3. DATASET, GROUND TRUTH AND EVALUATION METHODOLOGY

3.1 Dataset composition

Page group	Count assumed	Characteristics	Purpose
Light pages	2	Pale print, bright paper	Test gamma and threshold sensitivity
Normal pages	2	Even lighting, moderate contrast	Baseline comparison
Shadowed pages	2	One side darker	Test CLAHE, adaptive threshold and homomorphic filtering
Stained pages	2	Tea marks and ink blots	Test region segmentation
Noisy/striped pages	2	Dust, grain and horizontal bands	Test denoising and frequency filtering
Total	10	A4 grayscale pages	Minimum local dataset

3.2 Ground-truth creation

Ground-truth masks should be created by manually reviewing each page at 200–400% zoom. Separate masks are recommended for text, page boundary, ruled lines, dust, punch holes and stains. Annotators should follow a consistent policy: printed or handwritten text is foreground; paper texture is background; a stain is annotated as a defect region but not automatically classified as text. Save masks as single-channel PNG files with documented class values.

3.3 Evaluation formulas

Metric	Formula / interpretation
Dice similarity	Dice = 2TP / (2TP + FP + FN); 1.0 is perfect overlap.
Precision	TP / (TP + FP); high precision means fewer false detections.
Recall	TP / (TP + FN); high recall means fewer missed features.

IoU	TP / (TP + FP + FN); used for stain-region overlap.
PSNR	10 log10(MAX^2 / MSE); larger is generally better for denoising.
SSIM	Structural similarity of luminance, contrast and structure; closer to 1 is better.
CER	(substitutions + deletions + insertions) / reference characters.
WER	(word substitutions + deletions + insertions) / reference words.
Corner error	Mean Euclidean distance between estimated and labelled page corners.
Runtime	Elapsed wall-clock time for a complete page stage or pipeline.

3.4 Experimental protocol

1. Use identical source images and ground-truth masks for all candidate methods within a task.
2. Record software versions, CPU model, image dimensions and parameter values.
3. Warm up each method once, then time at least five repetitions and report the median.
4. Use the same OCR engine and language settings before and after enhancement.
5. Keep a copy of every output image, mask, histogram and log file.
6. Tune parameters on a development subset and report final results on held-out pages to reduce overfitting.

4. TASK 1 – BASIC GREY-LEVEL TRANSFORMATIONS

Assignment weight: 4 marks

Objective

Compare negative plus logarithmic transformation, power-law (gamma) transformation and piecewise-linear contrast stretching to make faded text darker and clearer.

Technical foundation and detailed explanation

Point processing maps each input intensity r to an output intensity s using s = T(r). It is computationally inexpensive because every output pixel depends only on the corresponding input pixel. For an 8-bit image, r and s lie in the range 0–255.

Negative transformation s = 255 − r reverses brightness. It can make dark ink visually bright on a dark background, but it does not improve the absolute separation between ink and paper. It is mainly useful for inspection and is not the preferred final scanner output.

Log transformation s = c log(1+r) expands darker values and compresses brighter values. The constant c is commonly selected as 255/log(256) to map the maximum input to 255. It can reveal low-intensity variations, but may compress paper differences and amplify noise in dark regions.

Power-law transformation s = c r^γ uses normalized intensity r/255. A gamma greater than one darkens mid-tones; gamma less than one brightens them. For faded dark text on a light paper background, gamma greater than one may increase apparent darkness, but the paper also becomes darker. A value that is too high can merge text with dark stains or clip details.

Piecewise-linear stretching maps a selected input interval to a larger output interval. A useful design is to map a paper/ink intensity band using measured control points rather than blindly stretching the entire 90–180 range. Grey-level slicing can highlight an intensity band, but can suppress faint pencil marks outside that band.

Assumed experimental parameters

Technique / setting	Illustrative value	Reason for selection
Negative/log	c = 255/log(256)	Display/diagnostic comparison
Gamma	γ = 1.35; normalized input	Moderate darkening of mid-tones
Piecewise	r1=90, r2=180; output	Expand useful intensity

	anchors 35 and 235	range

Comparative analysis

Candidate	Advantages	Limitations / risks	Relative runtime	Assessment
Negative + log	Contrast may improve for selected dark values	Can invert expected polarity or amplify dark noise	Low	Not selected for final output
Gamma	Adjustable mid-tone darkening	Sensitive to page brightness; may darken paper	Very low	Useful secondary option
Piecewise-linear	Direct control of selected band	Clipping can remove faint pencil detail	Very low	Selected if page calibration is available

Illustrative result table – NOT measured

The following values are assumed solely to demonstrate how the report should interpret results. They are not obtained by executing the code on the student's dataset and must be replaced with measured values.

Method	Status	Ink-paper difference (grey levels)	Time/page	Observation
Negative/log	Illustrative	12.4	0.7 ms	Inspection only
Gamma 1.35	Illustrative	31.8	0.5 ms	Good mid-tone separation
Piecewise stretch	Illustrative	38.6	0.6 ms	Best controlled contrast

Interpretation of results

The illustrative table shows piecewise stretching producing the largest mean ink-paper separation. This is plausible because the control points directly expand the intensity interval occupied by useful content. The conclusion must be confirmed by calculating the mean of manually annotated ink and paper pixels for all ten pages. A common gamma value is not universally suitable: a very light page may be over-darkened, while a very dark page may lose paper detail. Per-page calibration or a local method is safer.

Failure analysis and diagnostic cases

A large gamma can compress bright paper values and make a page appear grey. Aggressive stretching can clip faint pencil marks if their intensity falls outside the selected mapping interval. Negative transformation changes polarity and is unsuitable as the default OCR input unless inverted again.

Final decision and justification

Use piecewise-linear contrast stretching as the primary point transform when control points are estimated from page statistics. Use gamma correction as a low-cost alternative for pages with stable exposure. Keep the original image and avoid destructive clipping.

Implementation notes

Use cv2.LUT for lookup-table transforms. Create a 256-entry uint8 lookup table once and apply it to every pixel. Plot the transfer curve with matplotlib and calculate mean intensity separately over ink and paper masks. Preserve a copy of the original image to inspect clipping.

5. TASK 2 – HISTOGRAM PROCESSING

Assignment weight: 4 marks

Objective

Compare global histogram equalization, histogram matching and CLAHE for pages with different brightness and spatially varying shadows.

Technical foundation and detailed explanation

A histogram counts the number of pixels at each grey level. Global histogram equalization uses the cumulative distribution function to map intensities across the available range. It can improve global contrast, but it does not know whether a pixel belongs to text, paper or a stain.

Histogram matching maps the cumulative distribution of a source page to that of a clean reference page. It can standardize a collection, provided that the reference page is representative. If the reference is too bright or has a different paper/ink composition, matching may create undesirable tonal shifts.

CLAHE divides an image into tiles, equalizes each tile and limits histogram amplification using a clip limit. Interpolation between neighbouring tiles reduces visible tile boundaries. It is often effective for spatially varying illumination because each region is adjusted locally.

Global equalization can fail under a side shadow because a single global mapping is applied to both bright and dark regions. It may improve the overall histogram while leaving the local contrast in the shadowed area inadequate. CLAHE uses local distributions, but excessive clip limits can amplify camera grain, dust and paper texture.

Local mean and variance provide a useful diagnostic. A shadowed paper region has a lower local mean than a bright region. CLAHE can normalize local contrast, but a tile size that is too small may treat individual letters and stains as independent distributions.

Assumed experimental parameters

Technique / setting	Illustrative value	Reason for selection
Global equalization	cv2.equalizeHist	No tunable parameter
Histogram matching	Reference CDF from clean page P02	Reference must be representative
CLAHE	clipLimit=2.0; tileGridSize=(8,8)	Initial candidate; validate on data

Comparative analysis

Candidate	Advantages	Limitations / risks	Relative runtime	Assessment
Global equalization	Simple global contrast improvement	Weak shadow correction; may amplify blank-page texture	Very low	Baseline comparison
Histogram matching	Consistent appearance across a collection	Depends strongly on reference page	Low	Useful for standardized archives
CLAHE	Local contrast and shadow improvement	High clip limit amplifies noise; tile artefacts possible	Low	Selected for uneven lighting

Illustrative result table – NOT measured

The following values are assumed solely to demonstrate how the report should interpret results. They are not obtained by executing the code on the student's dataset and must be replaced with measured values.

Method	Status	Illustrative contrast	Entropy (bits)	Time/page
Global equalization	Illustrative	0.872	6.91	0.8 ms
Histogram matching	Illustrative	0.889	6.42	1.1 ms
CLAHE 2.0	Illustrative	0.914	6.55	2.0 ms

Interpretation of results

Illustrative contrast and entropy values suggest CLAHE can improve local text visibility while avoiding the extreme noise increase associated with unrestricted

equalization. Entropy alone is not a quality score: noise and stains also increase entropy. Inspect blank paper regions and calculate OCR or Dice performance before selecting parameters.

Failure analysis and diagnostic cases

Equalizing a page containing approximately 90% blank paper can make the dominant paper histogram control the mapping, leaving faint text insufficiently enhanced or exaggerating minor paper variations. A very high CLAHE clip limit can turn camera grain into high-contrast texture and cause thresholding to produce false foreground patches.

Final decision and justification

Select CLAHE with a moderate clip limit as the default for shadowed pages. Use histogram matching when many pages share the same print and paper characteristics. Retain the unmodified image for archival integrity.

Implementation notes

Use cv2.equalizeHist for global equalization and cv2.createCLAHE for CLAHE. Histogram matching can be implemented by computing source and reference CDFs, then mapping each source intensity to the nearest reference CDF value. Record clipLimit and tileGridSize in the experiment log.

6. TASK 3 – SPATIAL FILTERING AND IMAGE SMOOTHING

Assignment weight: 5 marks

Objective

Compare box, Gaussian and median filters for removing grain and dust without destroying 2–4 pixel text strokes.

Technical foundation and detailed explanation

The box filter replaces a pixel with the arithmetic mean of its neighbourhood. It reduces random grain but blurs edges and small punctuation. A 7×7 kernel covers a region wider than many text strokes, so it can make thin characters disappear or join nearby characters.

The Gaussian filter uses distance-weighted averaging. Nearby pixels contribute more than distant pixels. It produces smoother and more natural blur than a box filter, but it still softens thin strokes. A 3×3 Gaussian kernel is a reasonable starting point for small grain.

The median filter replaces each pixel by the median of its neighbourhood. It is nonlinear and particularly effective for salt-and-pepper noise because isolated extreme values are rejected. It generally preserves edges better than averaging filters, although a large median kernel can remove punctuation and small dots.

PSNR compares pixel-wise reconstruction error, while SSIM measures structural similarity. A filter can achieve a high PSNR by smoothing aggressively yet damage OCR-critical punctuation. Therefore, image metrics should be combined with text preservation and OCR measurements.

Correlation and convolution differ by kernel flipping. For symmetric averaging, Gaussian and median neighbourhoods, flipping does not change the result. For an asymmetric directional kernel, the orientation changes. OpenCV filter2D performs correlation by default; a manually flipped kernel is needed to implement convolution.

Assumed experimental parameters

Technique / setting	Illustrative value	Reason for selection
Box	3×3 and 5×5	Use 3×3 only for mild grain
Gaussian	3×3, σ=0.8	Gentle smoothing before edge detection
Median	3×3	Primary salt-and-pepper cleanup

Comparative analysis

Candidate	Advantages	Limitations / risks	Relative runtime	Assessment
Box	Random grain reduction	Softens all edges; dots and commas may vanish	Very low	Not preferred
Gaussian	Good smooth grain reduction	Some stroke blur	Low	Use lightly before edge detection
Median	Strong isolated speck removal	Large kernels remove small characters	Low	Selected for impulse noise

Illustrative result table – NOT measured

The following values are assumed solely to demonstrate how the report should interpret results. They are not obtained by executing the code on the student's dataset and must be replaced with measured values.

Method	Status	PSNR	SSIM	Time/page
Box 3×3	Illustrative	28.7 dB	0.901	0.35 ms
Gaussian 3×3	Illustrative	30.2 dB	0.925	0.55 ms
Median 3×3	Illustrative	32.1 dB	0.941	1.1 ms

Interpretation of results

The assumed example shows the median filter obtaining the strongest impulse-noise result, while Gaussian smoothing provides a balanced treatment of camera grain. In practice, use a noise-aware sequence: median filtering for isolated specks, followed by a very light Gaussian operation only if grain remains. Do not automatically apply both to every page.

Failure analysis and diagnostic cases

A 7×7 box filter can average a complete 2–4 pixel stroke with surrounding paper and reduce its contrast until it disappears. Omitting smoothing before edge detection can create many false edges from grain, dust and scanner stripes. Zero padding can create artificial dark borders; BORDER_REFLECT generally reduces this effect for document images.

Final decision and justification

Use 3×3 median for salt-and-pepper specks and 3×3 Gaussian with small sigma for camera grain. Choose one or both based on measured noise type; avoid 5×5 or 7×7 smoothing unless thin-text preservation is demonstrated.

Implementation notes

Use cv2.blur, cv2.GaussianBlur and cv2.medianBlur. Add known Gaussian and salt-and-pepper noise to clean pages, preserve the clean source as the reference and calculate PSNR/SSIM. For line profiles, sample a fixed row crossing representative characters before and after filtering.

7. TASK 4 – IMAGE SHARPENING AND EDGE ENHANCEMENT

Assignment weight: 5 marks

Objective

Compare Laplacian sharpening, unsharp masking/high-boost filtering and gradient-based sharpening for slightly blurred phone photographs.

Technical foundation and detailed explanation

The Laplacian is a second spatial derivative that responds to rapid intensity changes in all directions. A sharpened image can be formed as $g = f - c\nabla^2 f$. The sign and scale of the Laplacian kernel determine whether edges are added or subtracted. The derivative output may contain negative values and must be converted carefully for display.

Unsharp masking blurs the image, subtracts the blurred version from the original to obtain a detail mask, then adds a scaled mask back. The expression is $g = f + k(f - f_{blur})$. k=1 is conventional unsharp masking; larger k is high-boost filtering. It is intuitive and often gives controllable improvement.

Gradient sharpening uses first derivatives such as Sobel in x and y. Gradient magnitude indicates edge strength. Adding the magnitude can strengthen boundaries, but it may also make noise and stains more prominent. Directional information can be useful for diagnosing text orientation.

Second derivatives respond to changes in gradient and can emphasize fine transitions, but they are more sensitive to noise. First derivatives measure edge strength and are often less aggressive. A light smoothing step before derivative-based sharpening helps prevent grain from becoming exaggerated.

Variance of the Laplacian is a simple focus/sharpness indicator. It is not a complete quality metric: noisy images can have a high variance even when text is unreadable. Pair it with blank-paper noise standard deviation, OCR CER and visual inspection.

Assumed experimental parameters

Technique / setting	Illustrative value	Reason for selection
Laplacian	3×3 4-neighbour; c=0.35	Scale derivative output before display
Unsharp mask	Gaussian 3×3, σ=1.0, k=1.0	Moderate detail enhancement

Sobel gradient	ksize=3, gain=0.25	Use cautiously after smoothing

Comparative analysis

Candidate	Advantages	Limitations / risks	Relative runtime	Assessment
Laplacian	Strong fine edge emphasis	Noise amplification and halos	Low	Useful for analysis, not default
Unsharp masking	Controllable crispness	Large k creates halos and exaggerated stains	Low	Selected for mild blur
Gradient sharpening	Emphasizes directional edges	Can brighten all edges and noise	Low	Use for edge diagnostics

Illustrative result table – NOT measured

The following values are assumed solely to demonstrate how the report should interpret results. They are not obtained by executing the code on the student's dataset and must be replaced with measured values.

Method	Status	Laplacian variance	Blank-region noise σ	Observation
Laplacian	Illustrative	142.5	7.8	Moderate halos
Unsharp k=1	Illustrative	126.4	4.2	Best balance
Sobel gain .25	Illustrative	118.9	5.1	Edge emphasis

Interpretation of results

The example favours unsharp masking because it improves apparent sharpness while limiting blank-paper noise. The selection should be rejected if OCR CER worsens or if faint strokes become thicker than their original form. Sharpening is optional: a sharp source should not be sharpened again.

Failure analysis and diagnostic cases

Applying high-boost with a large k to an unsmoothed photograph increases grain, stain boundaries and JPEG artefacts. Displaying a signed Laplacian directly as uint8 clips negative values and loses information. Convert to a signed floating-point image, combine with the original, then clip only at the final display stage.

Final decision and justification

Use mild unsharp masking only when a focus measure and visual inspection indicate blur. Use k around 0.8–1.0 as an initial assumption and validate OCR. Do not sharpen already crisp scans.

Implementation notes

Use cv2.Laplacian with CV_32F, GaussianBlur and addWeighted. For Sobel, calculate dx and dy then cv2.magnitude. Measure sharpness only within text regions and noise only within manually selected blank-paper regions.

8. TASK 5 – FREQUENCY-DOMAIN LOW-PASS FILTERING

Assignment weight: 5 marks

Objective

Compare Ideal, Butterworth and Gaussian low-pass filters for camera grain and faint horizontal scanner stripes.

Technical foundation and detailed explanation

The two-dimensional DFT represents an image as spatial frequencies. Low frequencies describe slowly changing illumination and broad regions; high frequencies describe fine edges, text strokes and noise. A low-pass filter attenuates high-frequency components.

The Ideal Low-Pass Filter passes all frequencies inside a radius D0 and rejects all frequencies outside it. Its abrupt transition can cause ringing near sharp text edges because a sharp frequency cut-off corresponds to oscillatory behaviour in the spatial domain.

The Butterworth Low-Pass Filter uses an order n to control the transition slope. A higher order approaches an ideal filter and may increase ringing. A lower order provides a smoother transition but can blur more broadly.

The Gaussian Low-Pass Filter has a smooth frequency response and no abrupt cut-off. It generally avoids strong ringing, but it can soften fine text if D0 is too small. The cut-off radius must be selected by inspecting the spectrum and the text profile.

Horizontal stripes create directional frequency components. A log-magnitude spectrum can reveal peaks or bands corresponding to periodic row patterns. Filtering should target the stripe-related frequencies without suppressing frequencies needed for character strokes.

Assumed experimental parameters

Technique / setting	Illustrative value	Reason for selection
Ideal LPF	D0 = 30, 45, 60 pixels (illustrative)	Pad before DFT and crop after inverse transform
Butterworth	D0=45, order n=2	Moderate roll-off
Gaussian	D0=45	Smooth response; preferred starting point

Comparative analysis

Candidate	Advantages	Limitations / risks	Relative runtime	Assessment
Ideal LPF	Strong rejection outside cut-off	Ringing and edge ripples	Low/medium	Not selected
Butterworth LPF	Adjustable transition	Higher order may ring	Medium	Alternative
Gaussian LPF	Smooth attenuation	Can blur thin strokes	Medium	Selected for mild grain

Illustrative result table – NOT measured

The following values are assumed solely to demonstrate how the report should interpret results. They are not obtained by executing the code on the student's dataset and must be replaced with measured values.

Method	Status	Dice / contrast index	Ringing or noise index	Observation
Ideal D0=45	Illustrative	0.921	0.74	Visible ringing
Butterworth n=2	Illustrative	0.934	0.69	Moderate blur
Gaussian D0=45	Illustrative	0.942	0.62	Lowest ringing

Interpretation of results

The example favours Gaussian low-pass filtering for general grain reduction, but scanner stripes should be treated as a directional periodic artefact rather than removed by indiscriminately lowering all high frequencies. A narrow notch or directional suppression around measured spectral peaks may preserve more text detail.

Failure analysis and diagnostic cases

Without padding, the DFT assumes periodic image boundaries and opposite edges can interact, producing wrap-around artefacts. An Ideal filter's abrupt transition can

cause ringing around high-contrast text. Excessive smoothing reduces edge contrast and can increase OCR character substitutions.

Final decision and justification

Use Gaussian low-pass for mild broad-band noise only when spatial filtering is insufficient. For periodic scanner stripes, identify the corresponding spectral components and use a narrow directional notch or stripe-specific correction. Compare OCR before retaining the frequency stage.

Implementation notes

Use np.fft.fft2, fftshift, a centred radial distance matrix, multiplication by the transfer function and inverse FFT. Pad before transformation and crop back to the original dimensions. Save log(1+abs(F)) spectra and quantify edge overshoot near a high-contrast text boundary.

9. TASK 6 – FREQUENCY-DOMAIN HIGH-PASS FILTERING

Assignment weight: 4 marks

Objective

Compare Ideal/Butterworth high-pass filtering, Gaussian high-frequency emphasis and homomorphic filtering for text on shadowed pages.

Technical foundation and detailed explanation

High-pass filters attenuate low spatial frequencies and retain high-frequency detail. This can emphasize text edges, but a pure high-pass output removes average paper brightness and broad tonal information. It should not be used directly as a final document image.

An Ideal or Butterworth HPF can be obtained as Hhp = 1 − Hlp. The Butterworth version provides a smoother transition than the ideal form. The output is commonly signed or centred around a display offset, so its visual appearance must not be confused with a natural grayscale page.

Gaussian high-frequency emphasis retains a portion of low-frequency content while increasing high-frequency components. The form H = a + bHhp, with b greater than a, provides a compromise between paper tone and edge enhancement.

Homomorphic filtering models the image as illumination multiplied by reflectance: f(x,y)=i(x,y)r(x,y). Taking logarithms converts the product into a sum. A high-emphasis frequency filter suppresses slowly varying illumination while retaining reflectance changes such as text. Exponentiation returns to the image domain.

The low-frequency gain γL should be lower than the high-frequency gain γH. Interchanging these gains boosts illumination variation and suppresses text detail, which is the opposite of the intended operation. Parameters must be tuned to avoid flattening paper tone or exaggerating stains.

Assumed experimental parameters

Technique / setting	Illustrative value	Reason for selection
Ideal/Butterworth HPF	D0=25; Butterworth order 2	Display only after scaling/offset
Gaussian emphasis	a=0.7, b=1.5	Retain low-frequency paper tone
Homomorphic	γL=0.6, γH=1.5, D0=30	Initial shadow correction

Comparative analysis

Candidate	Advantages	Limitations / risks	Relative runtime	Assessment
Ideal/Butterworth HPF	Strong edge emphasis	Loses DC and paper brightness; ringing	Low/medium	Not final output
Gaussian emphasis	Preserves some base tone	May still amplify noise	Medium	Useful optional stage
Homomorphic	Corrects slow illumination variation	Parameter-sensitive; may amplify noise	Medium	Selected for strong shadows

Illustrative result table – NOT measured

The following values are assumed solely to demonstrate how the report should interpret results. They are not obtained by executing the code on the student's dataset and must be replaced with measured values.

Method	Status	Dice / contrast index	Ringing or noise index	Observation
Butterworth HPF	Illustrative	42.1	8.1	Paper tone not preserved
Gaussian emphasis	Illustrative	36.8	5.6	Good edge visibility
Homomorphic	Illustrative	39.4	4.7	Best shadow uniformity

Interpretation of results

The illustrative comparison indicates homomorphic filtering may provide the best shadow uniformity because it explicitly separates illumination and reflectance. However, it is not guaranteed to outperform CLAHE on every page. Evaluate text contrast, blank-paper standard deviation and OCR CER. A learned shadow-removal model may be tested, but a CPU-only deployment must account for inference time and possible stroke hallucination.

Failure analysis and diagnostic cases

An unmodified HPF output is not a valid final scan because it removes low-frequency paper tone and average brightness. Reversing γL and γH boosts the shadow field while weakening text detail. Over-aggressive high-frequency gain can make dust and JPEG artefacts prominent.

Final decision and justification

Use homomorphic filtering only on pages with substantial smooth illumination variation. For moderate shadows, CLAHE is simpler and faster. Do not apply pure HPF output to the final OCR or archival image.

Implementation notes

Use the logarithm of a positive-shifted image for homomorphic filtering. Construct a high-emphasis transfer function and exponentiate after inverse DFT. Compare the output with CLAHE and report blank-paper standard deviation, text contrast and OCR CER.

10. TASK 7 – POINT, LINE AND EDGE DETECTION

Assignment weight: 5 marks

Objective

Compare point/line masks, gradient operators and advanced Marr–Hildreth/Canny edge detection for dust, ruled lines and page borders.

Technical foundation and detailed explanation

Point detection uses a neighbourhood mask that responds strongly when a pixel differs from its neighbours. A threshold on the absolute response can identify isolated dark or bright specks. Text punctuation can also look point-like, so connected-component size, shape and location should be used to avoid false classification.

Line detection uses directional masks for 0°, 45°, 90° and −45°. These masks respond to aligned structures such as ruled lines and table borders. Slight page rotation can reduce the response because a fixed mask no longer aligns with the line. Deskewing or multi-angle morphology can improve robustness.

Roberts, Prewitt and Sobel estimate first derivatives. Gradient magnitude combines horizontal and vertical changes. Sobel includes a degree of smoothing and is often a practical baseline, but noise and text edges generate many responses.

Marr–Hildreth applies Gaussian smoothing followed by a Laplacian and detects zero crossings. Canny performs smoothing, gradient estimation, non-maximum suppression and hysteresis thresholding. Canny generally produces thin, connected edges when thresholds are appropriate.

Precision and recall should be measured separately for dust, ruled lines and page borders. A detector that produces many text edges may have high recall but poor precision for page boundaries. The page-border task should use geometric constraints such as four long sides and a plausible quadrilateral.

Assumed experimental parameters

Technique / setting	Illustrative value	Reason for selection
Point response	3×3 Laplacian; |R| threshold T=35	Illustrative threshold; tune from validation
Sobel	ksize=3; magnitude threshold	Use after light smoothing
Canny	low=50, high=150	Starting values; adapt to contrast

Hough	minimum line length 0.25 page width	Reject short text strokes

Comparative analysis

Candidate	Advantages	Limitations / risks	Relative runtime	Assessment
Point/line masks	Directly targets isolated points and orientations	Fixed orientation and text false positives	Low	Use for dust/ruled lines
Gradient operators	Simple edge magnitude and direction	Noise-sensitive; thick edges possible	Low	Useful intermediate
Canny/Marr–Hildreth	Thin and connected edge maps	Threshold-sensitive; Canny can break under shadows	Low/medium	Selected for page boundary

Illustrative result table – NOT measured

The following values are assumed solely to demonstrate how the report should interpret results. They are not obtained by executing the code on the student's dataset and must be replaced with measured values.

Method	Status	Precision	Recall	Observation
Point/line masks	Illustrative	0.88	0.82	Good for aligned rules
Sobel	Illustrative	0.84	0.91	More false edges
Canny	Illustrative	0.93	0.90	Best border continuity

Interpretation of results

Use different detectors for different feature classes rather than expecting one operator to detect everything. A point mask plus component filtering is suitable for candidate dust specks. Morphological horizontal/vertical extraction is suitable for ruled lines. Canny followed by Hough line grouping or contour approximation is suitable for page borders.

Failure analysis and diagnostic cases

Applying edge detection directly to noisy photographs creates false responses from grain, stains and JPEG blocks. Treating every detected character as a point causes punctuation and isolated letters to be incorrectly marked as dust. A text-aware exclusion mask and component geometry reduce this error.

Final decision and justification

Select Canny with hysteresis for page-border candidates, Hough/contour geometry for the four sides, morphology for ruled lines, and point-response plus component filtering for dust. Compare Canny with HED only if CPU runtime and model availability permit.

Implementation notes

Use filter2D for point and directional masks, Sobel for gradient magnitude, Canny for edges, HoughLinesP for line segments and findContours for closed shapes. Compare against hand annotations with a two-pixel matching tolerance.

11. TASK 8 – THRESHOLDING AND BINARIZATION

Assignment weight: 4 marks

Objective

Compare iterative global thresholding, Otsu global thresholding and adaptive thresholding for converting text and paper into binary classes.

Technical foundation and detailed explanation

Global thresholding assigns pixels below T to foreground and pixels above T to background, or vice versa depending on polarity. An iterative method begins with an initial threshold, computes the mean of the two classes, updates T to the average of the means and repeats until convergence.

Otsu thresholding searches for the threshold that maximizes between-class variance. It is effective when the histogram has separable foreground and background populations. It assumes one global threshold is suitable for the full image.

Adaptive thresholding calculates a local threshold from a neighbourhood. Mean or Gaussian-weighted local statistics allow the threshold to vary across the page. This is valuable when one side is shadowed and the other is bright.

Dice similarity compares the resulting text mask with a ground-truth mask. It is sensitive to both missing foreground and false foreground. The assignment target is Dice ≥ 0.90. A good visual result should also preserve character connectivity and avoid merging text with stains.

The illumination-reflectance model explains the limitation of global thresholding. If the illumination i(x,y) changes spatially, the same reflectance value can produce different observed grey levels. A single threshold may classify dark-side paper as text or fail to detect faded ink on the bright side.

Assumed experimental parameters

Technique / setting	Illustrative value	Reason for selection
Iterative global	Initial T = image mean; convergence tolerance 0.5	Baseline global method
Otsu	THRESH_BINARY_INV + OTSU	Invert according to image polarity
Adaptive Gaussian	blockSize=31; C=10	Odd block size larger than characters

Comparative analysis

Candidate	Advantages	Limitations / risks	Relative runtime	Assessment
Iterative global	Simple and interpretable	Fails under strong spatial shadow	Very low	Not selected for shadowed pages
Otsu	Automatic global threshold	Sensitive to stains and bimodal histogram assumption	Very low	Good for even scans
Adaptive	Local illumination compensation	Small block fragments characters; large block can miss local detail	Low	Selected for shadowed pages

Illustrative result table – NOT measured

The following values are assumed solely to demonstrate how the report should interpret results. They are not obtained by executing the code on the student's dataset and must be replaced with measured values.

Method	Status	Dice	Time/page (s)	Observation
Iterative global	Illustrative	0.861	0.12	Weak shadow preservation
Otsu	Illustrative	0.884	0.16	Good on evenly lit pages
Adaptive Gaussian	Illustrative	0.923	0.24	Best shadowed-page Dice

Interpretation of results

The example values indicate adaptive Gaussian thresholding is the best general choice for uneven illumination. Otsu remains useful for flatbed scans with clean bimodal histograms. Gaussian smoothing before Otsu can reduce isolated noise, but excessive smoothing may weaken punctuation. The improvement must be measured rather than assumed.

Failure analysis and diagnostic cases

Otsu may classify the dark shadow as foreground and produce large black patches. An adaptive block size smaller than character dimensions responds to individual stroke fragments and may break characters. Very large blocks behave more like global thresholding and cannot follow local shadows.

Final decision and justification

Use adaptive Gaussian thresholding for shadowed photographs, with a block size larger than typical character height and a tuned C value. Use Otsu for evenly illuminated scans. Always compare Dice and OCR output.

Implementation notes

Use threshold, THRESH_OTSU and adaptiveThreshold. Ensure the output polarity is consistent with the ground-truth mask. Evaluate Dice on each of at least ten pages and report light, normal, shadowed and stained subsets separately.

12. TASK 9 – REGION SEGMENTATION AND INTEGRATED PIPELINE

Assignment weight: 6 marks

Objective

Compare region growing, region split-and-merge and edge linking/boundary detection, then integrate selected stages into a smart-scanner pipeline.

Technical foundation and detailed explanation

Region growing starts from a seed pixel and adds neighbouring pixels whose intensity or colour is similar to the seed. It is intuitive for stains with relatively uniform appearance. Its weakness is seed dependence: a seed placed on text can grow through connected character pixels or leak into nearby regions.

Region splitting and merging recursively divides an image into quadrants until each region is sufficiently uniform, then merges neighbouring regions with similar statistics. It can represent broad stains and paper areas, but may be computationally expensive and sensitive to the homogeneity threshold.

Edge linking connects edge pixels based on strength, orientation and proximity. Hough transforms detect long straight lines, while contours trace connected boundaries. This is suitable for page borders and ruled lines, especially when geometric constraints are added.

Stain-region IoU measures overlap between a predicted stain region and the annotated region. Page-corner error measures the distance between predicted and labelled corners. Boundary gaps can be bridged with morphological closing or line fitting, but closing must not join unrelated text components.

The integrated scanner should not treat all defects as one class. Dust specks are small isolated components; punch holes are usually larger, near-circular regions; stains are diffuse connected regions; ink blots may be dark and irregular; page boundaries are long geometric edges.

Assumed experimental parameters

Technique / setting	Illustrative value	Reason for selection
Region growing	8-connectivity; intensity tolerance ±12	Seed from staff-confirmed candidate stain
Split/merge	Minimum block 16×16; local variance threshold	Illustrative settings
Edge linking	Canny 50/150; HoughP line grouping	Use page geometry and contour closure

Comparative analysis

Candidate	Advantages	Limitations / risks	Relative runtime	Assessment
Region growing	Can capture connected stain shapes	Seed-dependent leakage into text	Low/medium	Use for stain candidate refinement
Split/merge	Handles broad homogeneous regions	More complex and block artefacts	Medium/high	Not selected for baseline
Edge linking/Hough	Strong page-border and line geometry	Interrupted edges under shadows	Low/medium	Selected for page crop and table lines

Illustrative result table – NOT measured

The following values are assumed solely to demonstrate how the report should interpret results. They are not obtained by executing the code on the student's dataset and must be replaced with measured values.

Method	Status	Stain IoU	Corner error	Observation
Region growing	Illustrative	0.78 IoU	5.8 px corner error	Sensitive to seed
Split/merge	Illustrative	0.81 IoU	4.9 px	Higher runtime
Edge linking	Illustrative	0.74 stain IoU	2.6 px	Best corner localization

Interpretation of results

The example shows why the best method is feature-dependent. Region growing has value for stain masks, while edge linking provides the most reliable page geometry. A hybrid approach is therefore stronger than choosing a single method for every output. Staff review remains appropriate for uncertain defect annotations.

Failure analysis and diagnostic cases

A seed on a printed character can cause region growing to include text and connected background variations. A page boundary interrupted by a shadow may

produce separate edge fragments; Hough grouping, quadrilateral constraints, morphology and corner geometry can restore a plausible closed boundary.

Final decision and justification

Use edge linking, Hough lines and contours for page-border localization; use region growing or connected-component methods for stain candidates; retain manual review for ambiguous stains and punch holes. The complete original pipeline should be measured against the 3-second target.

Implementation notes

Use floodFill or a queue-based 8-connected region-growing implementation for stain candidates. Use Hough line grouping for page sides, order corners consistently, and use getPerspectiveTransform/warpPerspective for cropping. Time every stage separately.

13. TASK 10 – RE-APPLY AND ADAPT THE SMART SCANNER PIPELINE

Assignment weight: 8 marks

Objective

Adapt the original grayscale pipeline to 12-megapixel 4000×3000 colour JPEG phone photographs, angled pages, blue pen annotations and a 1-second processing limit.

Technical foundation and detailed explanation

The changed input invalidates several assumptions from the original task. The image is colour rather than grayscale, the pixel count increases to 12 megapixels, JPEG compression introduces 8×8 block patterns, page geometry may be perspective-distorted, classroom lighting can be uneven, and blue pen notes must be preserved.

Retaining the original pipeline unchanged is unlikely to meet the one-second limit because it processes a larger image and discards colour information. Grayscale conversion can make light blue notes weak or indistinguishable from the paper. The original A4 scan assumption also does not address page tilt and perspective.

The MODIFY strategy resizes the image to a working resolution, tunes CLAHE and filters, and selects a colour channel where blue ink has strong contrast. Blue ink often appears darker in a yellow/blue-sensitive channel relationship, but the correct channel should be measured for the actual camera and lighting. HSV or Lab colour representations may help separate chromatic annotations.

The REDESIGN strategy uses homomorphic illumination correction, frequency-domain treatment for JPEG blocks, Hough-based page detection and perspective correction. These steps can be useful but may exceed the one-second CPU budget if applied to every full-resolution image. JPEG block artefacts are often better handled with codec-aware denoising or carefully selected spatial filtering rather than broad frequency suppression.

Perspective correction estimates four page corners and maps them to a rectangular destination using cv2.getPerspectiveTransform and cv2.warpPerspective. A low-resolution copy can be used to detect corners; the transform can then be applied to the original or a moderately resized image.

Assumed experimental parameters

Technique / setting	Illustrative value	Reason for selection
Strategy A – Retain	Original pipeline unchanged	Expected to fail colour annotation and time constraints
Strategy B – Modify	Resize to 1600 px long side; selective channel; CLAHE	Recommended practical compromise
Strategy C – Redesign	Homomorphic + frequency + Hough + perspective	Use when severe geometry/illumination requires it

Comparative analysis

Candidate	Advantages	Limitations / risks	Relative runtime	Assessment
A – Retain	No redesign effort	Slow; may lose blue pen notes; no perspective correction	Low	Reject
B – Modify	Balances speed and preservation	Some fine detail may be lost at reduced scale	Low/medium	Selected
C – Redesign	Best geometric/illumination capability	More complex and potentially slower than 1 s	High	Conditional fallback

Illustrative result table – NOT measured

The following values are assumed solely to demonstrate how the report should interpret results. They are not obtained by executing the code on the student's dataset and must be replaced with measured values.

Strategy	Status	Dice	Time/page	Blue-note recall
A – Retain	Illustrative	0.886	0.74 s	Blue annotation recall 0.62
B – Modify	Illustrative	0.918	0.82 s	Blue annotation recall 0.95
C – Redesign	Illustrative	0.927	1.46 s	Best corners, exceeds target

Interpretation of results

The assumed values support MODIFY as the best compromise. Strategy C may give slightly higher Dice and better perspective correction, but it misses the one-second requirement. Strategy B preserves blue notes by retaining a suitable colour channel and reduces computational load through resizing. The exact resize scale and processing time must be tested on at least 15 phone photographs.

Failure analysis and diagnostic cases

Applying the original grayscale pipeline can erase light blue marginal notes, blur JPEG-compressed characters and miss page corners when the page is angled. Processing all 12 megapixels through every filter can exceed the time limit. A reduced-resolution working copy should be used for detection, with selective full-resolution operations only where necessary.

Final decision and justification

Choose MODIFY. Detect page corners on a resized image, preserve colour data until blue-note extraction, apply perspective correction, use moderate local contrast enhancement and adaptive thresholding for the text layer, and retain the original colour crop as the archival record. Use document-AI models for layout interpretation/OCR assistance, not as a replacement for geometric correction and pixel-level defect annotations.

Implementation notes

Keep the original colour image. Create a resized working copy for page detection and segmentation. Evaluate candidate colour channels for blue-note contrast. Apply the resulting geometric transform to the colour image, then create separate outputs for text binarization, annotations and archival colour crop.

14. OVERALL RESULTS AND DESIGN DECISIONS

14.1 Consolidated method selection

Pipeline stage	Selected method	Initial parameter	Reason	Fallback / caution
Point enhancement	Piecewise-linear stretch	Calibrated control points	Directly expands useful band	Avoid clipping pencil marks
Histogram enhancement	CLAHE	clipLimit 2.0; 8×8 tiles	Local shadow compensation	Lower clip limit if paper noise rises
Impulse denoising	Median filter	3×3	Removes isolated specks	Protect punctuation
Grain smoothing	Gaussian	3×3, $\sigma \approx 0.8$	Gentle noise reduction	Skip if text is already sharp
Sharpening	Unsharp masking	$k \approx 0.8–1.0$	Controllable detail gain	Skip if halos/OCR errors increase
Periodic stripes	Directional frequency notch	Peak-specific	Targets periodic scanner pattern	Do not use broad LPF blindly
Page boundary	Canny + Hough + contours	50/150 starting thresholds	Long geometric edges	Use perspective constraints
Ruled lines	Morphological directional masks	Horizontal/vertical	Separates table rules	Deskew first
Binarization	Adaptive Gaussian	31 block; C=10	Handles spatial shadow	Otsu for flat scans
Stains	Region growing +	Tolerance tuned per page	Region-level annotations	Staff review for

	components			ambiguity
Phone adaptation	Modify strategy	Resize + colour preservation	Meets speed and blue-note constraints	Fallback to redesign if perspective is severe

14.2 Original 3-second processing budget

Stage	Illustrative time (ms)	Cumulative (ms)
Image read and grayscale	90	90
Median/Gaussian denoise	180	270
CLAHE / contrast	260	530
Optional unsharp mask	100	630
Canny and Hough	210	840
Adaptive threshold	330	1170
Connected components and annotations	260	1430
Crop, save and report	320	1750
Total illustrative time	1750 ms	1.75 s

The assumed 1.75-second total is below the 3-second target, leaving approximately 1.25 seconds of margin. This is a design estimate only. Actual timing must be collected on the specified lab PC. Neural inference, large image copies, disk I/O and OCR may increase total runtime and should be timed separately.

14.3 Design decision table

Decision	Evidence-based justification	Consequence if not selected
Use CLAHE on shadowed pages	Local contrast adapts to spatially varying illumination	Global methods may leave one side unreadable
Use small-kernel median	Impulse noise is present and text is only 2–4 px wide	Large averaging kernels can erase punctuation
Use sharpening conditionally	Some phone photos are slightly blurred	Unnecessary sharpening can amplify noise

Use adaptive thresholding for shadows	Local threshold responds to local background level	One global threshold may create black shadow patches
Use geometric page constraints	Page must be cropped as a quadrilateral	Raw edges may include text and background objects
Separate defect classes	Dust, punch holes and stains have different morphology	One detector produces high false-positive rates
Modify for phone uploads	Colour notes, perspective and 12 MP change assumptions	Retaining old pipeline loses annotations and time

14.4 Verification against predictions

The predictions across Tasks 1–8 should be checked against the final measurement sheet. If CLAHE improves Dice but worsens OCR CER, the likely cause is noise amplification or local threshold instability. If median filtering improves PSNR but punctuation recall falls, the kernel is too large or the source noise is not impulse-dominated. If a Gaussian frequency filter reduces stripes but also reduces text contrast, its cut-off is too low or the stripe components should be handled with a narrow notch.

14.5 Limitations

- Only ten original pages are required for the baseline study; this is a small sample and may not represent all library documents.
- Assumed numerical values in this draft are not experimental evidence.
- Manual ground truth can vary between annotators, especially for faint pencil strokes and diffuse stains.
- OCR performance depends on language, font, scan resolution and OCR engine configuration.
- A CPU-only system limits the practicality of large deep-learning restoration models.
- A stain can overlap text; preserving the original image and marking uncertain regions is safer than aggressive removal.

15. CONCLUSION AND FUTURE ENHANCEMENTS

The assignment demonstrates that smart document scanning requires a sequence of complementary image-processing operations rather than one universal filter. Point transformations are fast and useful for controlled contrast changes. CLAHE is a practical method for uneven illumination. Median and Gaussian filters address different noise types. Unsharp masking can improve mildly blurred text, but sharpening must be conditional. Frequency-domain filters are useful for periodic artefacts and illumination analysis, yet careless low-pass or high-pass filtering can damage character strokes or paper tone.

For segmentation, the most suitable method depends on the feature. Canny with geometric line grouping is appropriate for page borders; directional morphology is effective for ruled lines; point-response and connected-component analysis can identify dust candidates; and region growing can refine stain regions when seeds are reliable. Adaptive Gaussian thresholding is a strong candidate for shadowed pages, while Otsu remains useful for evenly illuminated scans.

The original scanner should use a modular hybrid pipeline and verify Dice ≥ 0.90 and runtime ≤ 3 seconds using the real dataset. Under the changed phone-photo requirement, MODIFY is recommended: work on a resized image for speed, retain colour to protect blue pen annotations, detect page corners and apply perspective correction, then perform selective enhancement and thresholding. A document-AI model may assist OCR and layout interpretation, but conventional image processing remains necessary for geometric correction and explicit defect annotations.

Future enhancements

- Add automatic blur and illumination quality checks to choose processing stages adaptively.
- Use a lightweight document dewarping model only when page curvature is detected.
- Add a staff review interface to accept, reject or correct dust and stain annotations.
- Store the original image, enhanced image, binary text mask and defect overlay as separate archival assets.
- Use OCR confidence to flag pages requiring manual inspection.
- Expand testing to multiple paper types, languages, handwriting styles and camera devices.

APPENDIX A – PYTHON IMPLEMENTATION FRAMEWORK

The following framework is intended as a starting point. It is deliberately written with standard OpenCV and NumPy operations. Adapt file paths, ground-truth naming and parameter values to the actual dataset. Execute the code locally and save logs; do not copy the illustrative tables as measured results.

```
import cv2
import numpy as np
import time
from pathlib import Path

def read_gray(path):
    image = cv2.imread(str(path), cv2.IMREAD_GRAYSCALE)
    if image is None:
        raise FileNotFoundError(path)
    return image

def gamma_lut(gamma):
    x = np.arange(256, dtype=np.float32) / 255.0
    y = np.clip((x ** gamma) * 255.0, 0, 255)
    return y.astype(np.uint8)

def apply_gamma(image, gamma=1.35):
    return cv2.LUT(image, gamma_lut(gamma))

def piecewise_stretch(image, r1=90, r2=180, s1=35, s2=235):
    lut = np.zeros(256, dtype=np.uint8)
    for r in range(256):
        if r <= r1:
            value = (s1 / max(r1, 1)) * r
        elif r <= r2:
            value = s1 + (s2-s1) * (r-r1) / (r2-r1)
        else:
            value = s2 + (255-s2) * (r-r2) / (255-r2)
        lut[r] = np.uint8(np.clip(value, 0, 255))
    return cv2.LUT(image, lut)

def histogram_match(source, reference):
    src_hist = np.bincount(source.ravel(), minlength=256)
    ref_hist = np.bincount(reference.ravel(), minlength=256)
```

```
    src_cdf = np.cumsum(src_hist).astype(np.float64)
    ref_cdf = np.cumsum(ref_hist).astype(np.float64)
    src_cdf /= src_cdf[-1]
    ref_cdf /= ref_cdf[-1]
    mapping = np.interp(src_cdf, ref_cdf, np.arange(256))
    return np.clip(mapping[source], 0, 255).astype(np.uint8)

def enhance_histograms(image):
    global_eq = cv2.equalizeHist(image)
    clahe = cv2.createCLAHE(clipLimit=2.0,
                            tileGridSize=(8, 8)).apply(image)
    return global_eq, clahe

def denoise(image):
    box = cv2.blur(image, (3, 3))
    gaussian = cv2.GaussianBlur(image, (3, 3), 0.8)
    median = cv2.medianBlur(image, 3)
    return box, gaussian, median

def unsharp(image, amount=1.0):
    blur = cv2.GaussianBlur(image, (3, 3), 1.0)
    return cv2.addWeighted(image, 1.0 + amount,
                           blur, -amount, 0)

def laplacian_sharpen(image, amount=0.35):
    lap = cv2.Laplacian(image, cv2.CV_32F, ksize=3)
    out = image.astype(np.float32) - amount * lap
    return np.clip(out, 0, 255).astype(np.uint8)

def threshold_methods(image):
    _, otsu = cv2.threshold(
        image, 0, 255, cv2.THRESH_BINARY_INV + cv2.THRESH_OTSU)
    adaptive = cv2.adaptiveThreshold(
        image, 255, cv2.ADAPTIVE_THRESH_GAUSSIAN_C,
        cv2.THRESH_BINARY_INV, 31, 10)
    return otsu, adaptive

def dice_score(pred, truth):
    pred = pred > 0
    truth = truth > 0
```

```
    denom = pred.sum() + truth.sum()
    if denom == 0:
        return 1.0
    return 2.0 * np.logical_and(pred, truth).sum() / denom

def psnr(reference, test):
    mse = np.mean((reference.astype(np.float32) -
                   test.astype(np.float32)) ** 2)
    if mse == 0:
        return float("inf")
    return 10 * np.log10((255.0 ** 2) / mse)

def timed_call(function, image, repeats=5):
    times = []
    result = None
    for _ in range(repeats):
        start = time.perf_counter()
        result = function(image)
        times.append(time.perf_counter() - start)
    return result, float(np.median(times))

def process_page(path, output_dir):
    image = read_gray(path)
    denoised = cv2.medianBlur(image, 3)
    enhanced = cv2.createCLAHE(
        clipLimit=2.0, tileGridSize=(8, 8)).apply(denoised)
    binary = cv2.adaptiveThreshold(
        enhanced, 255, cv2.ADAPTIVE_THRESH_GAUSSIAN_C,
        cv2.THRESH_BINARY_INV, 31, 10)
    edges = cv2.Canny(enhanced, 50, 150)

    output_dir = Path(output_dir)
    output_dir.mkdir(parents=True, exist_ok=True)
    stem = Path(path).stem
    cv2.imwrite(str(output_dir / f"{stem}_enhanced.png"), enhanced)
    cv2.imwrite(str(output_dir / f"{stem}_binary.png"), binary)
    cv2.imwrite(str(output_dir / f"{stem}_edges.png"), edges)
    return enhanced, binary, edges

if __name__ == "__main__":
```

```
# Replace these paths with the student's own dataset.
input_dir = Path("dataset/images")
output_dir = Path("outputs")
for image_path in sorted(input_dir.glob("*.png")):
    t0 = time.perf_counter()
    process_page(image_path, output_dir)
    elapsed = time.perf_counter() - t0
    print(image_path.name, f"{elapsed:.3f} seconds")
```

APPENDIX A – FREQUENCY-DOMAIN FILTERS AND GEOMETRY

```
import cv2
import numpy as np

def frequency_filter(gray, kind="gaussian", d0=45, order=2):
    rows, cols = gray.shape
    # Reflect padding reduces periodic boundary discontinuities.
    pad = cv2.copyMakeBorder(gray, rows//2, rows//2,
                             cols//2, cols//2,
                             cv2.BORDER_REFLECT)
    f = np.fft.fftshift(np.fft.fft2(pad))
    r, c = pad.shape
    yy, xx = np.ogrid[:r, :c]
    cy, cx = r // 2, c // 2
    distance = np.sqrt((yy-cy)**2 + (xx-cx)**2)

    if kind == "ideal":
        H = (distance <= d0).astype(np.float32)
    elif kind == "butterworth":
        H = 1.0 / (1.0 + (distance / d0)**(2*order))
    elif kind == "gaussian":
        H = np.exp(-(distance**2) / (2.0*d0*d0))
    else:
        raise ValueError("Unknown filter type")

    filtered = np.fft.ifft2(np.fft.ifftshift(f * H))
    filtered = np.real(filtered)
    filtered = filtered[rows//2:rows//2+rows,
                        cols//2:cols//2+cols]
    return np.clip(filtered, 0, 255).astype(np.uint8)

def order_points(points):
    points = np.asarray(points, dtype=np.float32)
    s = points.sum(axis=1)
    d = np.diff(points, axis=1).ravel()
    tl = points[np.argmin(s)]
    br = points[np.argmax(s)]
    tr = points[np.argmin(d)]
    bl = points[np.argmax(d)]
    return np.array([tl, tr, br, bl], dtype=np.float32)
```

```
def perspective_crop(image, corners):
    rect = order_points(corners)
    tl, tr, br, bl = rect
    width = int(max(np.linalg.norm(br-bl), np.linalg.norm(tr-tl)))
    height = int(max(np.linalg.norm(tr-br), np.linalg.norm(tl-bl)))
    destination = np.array([[0, 0], [width-1, 0],
                            [width-1, height-1],
                            [0, height-1]], dtype=np.float32)
    matrix = cv2.getPerspectiveTransform(rect, destination)
    return cv2.warpPerspective(image, matrix, (width, height))

def find_page_contour(gray):
    blur = cv2.GaussianBlur(gray, (5, 5), 0)
    edges = cv2.Canny(blur, 50, 150)
    contours, _ = cv2.findContours(
        edges, cv2.RETR_EXTERNAL, cv2.CHAIN_APPROX_SIMPLE)
    contours = sorted(contours, key=cv2.contourArea, reverse=True)
    for contour in contours[:10]:
        perimeter = cv2.arcLength(contour, True)
        approx = cv2.approxPolyDP(contour, 0.02*perimeter, True)
        if len(approx) == 4 and cv2.contourArea(approx) > 0.2*gray.size:
            return approx.reshape(4, 2)
    return None
```

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7. N. Otsu, “A threshold selection method from gray-level histograms,” IEEE Transactions on Systems, Man, and Cybernetics, 1979.

8. J. Canny, “A computational approach to edge detection,” IEEE Transactions on Pattern Analysis and Machine Intelligence, 1986.

9. R. C. Gonzalez, R. E. Woods and S. L. Eddins, Digital Image Processing Using MATLAB, McGraw-Hill.

10. Tesseract OCR documentation: https://tesseract-ocr.github.io/

11. OpenCV geometric image transformations documentation: https://docs.opencv.org/4.x/da/d6e/tutorial_py_geometric_transformations.html

FINAL SUBMISSION CHECKLIST

- Replace all illustrative assumed metrics with actual measurements from the selected dataset.
- Insert the 10 original page images and representative before/after outputs.
- Add histograms, CDF plots, transformation curves, spectra and intensity profiles generated by code.
- Add hand-annotated ground-truth masks and document the annotation method.
- Run all candidate methods on the same pages and record median runtime.
- Verify text Dice ≥ 0.90 and original pipeline runtime ≤ 3 seconds/page.
- For Task 10, test at least 15 colour phone photographs and verify the 1-second limit.
- Check that blue handwritten annotations are preserved.
- Confirm the Word document contains the correct student details and institution formatting.
- Review citations and ensure all external models/datasets used are named and acknowledged.

EXPERIMENTAL WORKSHEET – TASK 1

Record the input page condition, chosen control points, transformation parameters, ink mask, paper mask, mean ink value, mean paper value and their difference. Repeat for all ten pages. Include a separate note on faint pencil strokes and identify whether any input values were clipped at zero or 255.

Required observations

- Input image ID and acquisition condition.
- Method name, software version and all parameters.
- Before/after images saved at the same display scale.
- Numerical measurements and the script or log that generated them.
- Observed failure cases and explanation.
- Decision on whether the method should be retained in the final scanner.

Interpretation template

The method was evaluated on the same source images as the alternative techniques. Its main advantage was _____. Its main limitation was _____. The measured quality score was _____ and the median processing time was _____. Based on the accuracy, text preservation and runtime requirements, the method is recommended / not recommended because _____.

Evidence to attach

Attach the input image, output image, histogram or frequency plot where relevant, ground-truth overlay, metric table and execution log. Every figure should include a figure number, descriptive caption, page identifier and parameter settings.

EXPERIMENTAL WORKSHEET – TASK 2

For each representative light, normal and shadowed page, record the histogram, CDF, global mean, standard deviation, entropy, local mean and local variance. Compare equalization, matching and CLAHE. Inspect blank paper tiles for amplified grain and visible tile boundaries.

Required observations

- Input image ID and acquisition condition.
- Method name, software version and all parameters.
- Before/after images saved at the same display scale.
- Numerical measurements and the script or log that generated them.
- Observed failure cases and explanation.
- Decision on whether the method should be retained in the final scanner.

Interpretation template

The method was evaluated on the same source images as the alternative techniques. Its main advantage was _____. Its main limitation was _____. The measured quality score was _____ and the median processing time was _____. Based on the accuracy, text preservation and runtime requirements, the method is recommended / not recommended because _____.

Evidence to attach

Attach the input image, output image, histogram or frequency plot where relevant, ground-truth overlay, metric table and execution log. Every figure should include a figure number, descriptive caption, page identifier and parameter settings.

EXPERIMENTAL WORKSHEET – TASK 3

Prepare clean source pages and introduce Gaussian noise with sigma 6 plus salt-and-pepper noise at 1%. Compare 3×3, 5×5 and 7×7 kernels. Record PSNR, SSIM, punctuation preservation and line-profile width. Explain any metric that disagrees with visual/OCR quality.

Required observations

- Input image ID and acquisition condition.
- Method name, software version and all parameters.
- Before/after images saved at the same display scale.
- Numerical measurements and the script or log that generated them.
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Evidence to attach

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EXPERIMENTAL WORKSHEET – TASK 4

Select at least five slightly blurred phone images. Compare Laplacian, unsharp masking and Sobel-based enhancement. Measure variance of Laplacian in text regions and standard deviation in blank paper. Run OCR using the same engine and report CER.

Required observations

- Input image ID and acquisition condition.
- Method name, software version and all parameters.
- Before/after images saved at the same display scale.
- Numerical measurements and the script or log that generated them.
- Observed failure cases and explanation.
- Decision on whether the method should be retained in the final scanner.

Interpretation template

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Evidence to attach

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EXPERIMENTAL WORKSHEET – TASK 5

Save the centred log-magnitude spectrum for a striped scan. Mark the spectral components suspected to represent row stripes. Test three cut-off radii for each low-pass filter. Measure ringing near a sharp text edge and OCR change.

Required observations

- Input image ID and acquisition condition.
- Method name, software version and all parameters.
- Before/after images saved at the same display scale.
- Numerical measurements and the script or log that generated them.
- Observed failure cases and explanation.
- Decision on whether the method should be retained in the final scanner.

Interpretation template

The method was evaluated on the same source images as the alternative techniques. Its main advantage was _____. Its main limitation was _____. The measured quality score was _____ and the median processing time was _____. Based on the accuracy, text preservation and runtime requirements, the method is recommended / not recommended because _____.

Evidence to attach

Attach the input image, output image, histogram or frequency plot where relevant, ground-truth overlay, metric table and execution log. Every figure should include a figure number, descriptive caption, page identifier and parameter settings.

EXPERIMENTAL WORKSHEET – TASK 6

Select shadowed and evenly illuminated pages. Compare high-pass, high-frequency emphasis and homomorphic outputs. Record text-to-paper contrast, blank-paper uniformity, average brightness and OCR error. Explain why a pure high-pass image is not a final scan.

Required observations

- Input image ID and acquisition condition.
- Method name, software version and all parameters.
- Before/after images saved at the same display scale.
- Numerical measurements and the script or log that generated them.
- Observed failure cases and explanation.
- Decision on whether the method should be retained in the final scanner.

Interpretation template

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Evidence to attach

Attach the input image, output image, histogram or frequency plot where relevant, ground-truth overlay, metric table and execution log. Every figure should include a figure number, descriptive caption, page identifier and parameter settings.

EXPERIMENTAL WORKSHEET – TASK 7

Create separate reference masks for dust, ruled lines and page borders. Match detections within two pixels. Report precision and recall per feature. Record threshold values and explain the impact of slight page rotation.

Required observations

- Input image ID and acquisition condition.
- Method name, software version and all parameters.
- Before/after images saved at the same display scale.
- Numerical measurements and the script or log that generated them.
- Observed failure cases and explanation.
- Decision on whether the method should be retained in the final scanner.

Interpretation template

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Evidence to attach

Attach the input image, output image, histogram or frequency plot where relevant, ground-truth overlay, metric table and execution log. Every figure should include a figure number, descriptive caption, page identifier and parameter settings.

EXPERIMENTAL WORKSHEET – TASK 8

For all ten pages, save the iterative threshold, Otsu threshold and adaptive output. Record the selected global threshold, adaptive block size and C value. Calculate Dice and inspect black patches, broken characters and merged components.

Required observations

- Input image ID and acquisition condition.
- Method name, software version and all parameters.
- Before/after images saved at the same display scale.
- Numerical measurements and the script or log that generated them.
- Observed failure cases and explanation.
- Decision on whether the method should be retained in the final scanner.

Evidence to attach

Attach the input image, output image, histogram or frequency plot where relevant, ground-truth overlay, metric table and execution log. Every figure should include a figure number, descriptive caption, page identifier and parameter settings.

EXPERIMENTAL WORKSHEET – TASK 9

For each stain, record seed coordinates, intensity tolerance, connected region size and IoU. For each page, record four corner coordinates, mean corner error and time per stage. Compare region growing, split-and-merge and edge linking.

Required observations

- Input image ID and acquisition condition.
- Method name, software version and all parameters.
- Before/after images saved at the same display scale.
- Numerical measurements and the script or log that generated them.
- Observed failure cases and explanation.
- Decision on whether the method should be retained in the final scanner.

Evidence to attach

Attach the input image, output image, histogram or frequency plot where relevant, ground-truth overlay, metric table and execution log. Every figure should include a figure number, descriptive caption, page identifier and parameter settings.

EXPERIMENTAL WORKSHEET – TASK 10

Collect at least 15 phone photos under classroom lighting. Record image size, page angle, shadow severity, JPEG quality estimate, blue annotation presence, strategy output, Dice, corner error, blue-note recall and runtime.

Required observations

- Input image ID and acquisition condition.
- Method name, software version and all parameters.
- Before/after images saved at the same display scale.
- Numerical measurements and the script or log that generated them.
- Observed failure cases and explanation.
- Decision on whether the method should be retained in the final scanner.

Evidence to attach

Attach the input image, output image, histogram or frequency plot where relevant, ground-truth overlay, metric table and execution log. Every figure should include a figure number, descriptive caption, page identifier and parameter settings.

Parameter sensitivity and tuning strategy

Parameters should be tuned on a development subset and frozen before final evaluation. For gamma, sweep a small range such as 1.0, 1.15, 1.3 and 1.45. For CLAHE, compare clip limits such as 1.5, 2.0 and 3.0. For adaptive thresholding, test several odd block sizes around the character scale and small C changes. Select settings using Dice and OCR, not visual contrast alone.

Practical design implications

The implementation should use a conservative, reversible approach. Every enhancement stage should produce a separate output so that staff can compare it with the original. If a method improves one metric while damaging another, record the trade-off and select the stage based on the primary requirement of readable, faithful document preservation.

Validation questions

- Does the operation preserve all visible text and punctuation?
- Does it improve performance on shadowed and stained pages, not only clean scans?
- Does it introduce halos, false regions or artificial strokes?
- Does the method meet the per-page processing budget on the target CPU?
- Can the result be reproduced from saved parameters and source data?

OCR integration and text integrity

OCR is downstream of enhancement and segmentation. It should not be used as the only quality measure because language models can correct plausible words despite damaged pixels. Save raw OCR output and confidence, compare against verified reference text, and calculate CER/WER. Any method that invents, deletes or alters strokes should be rejected even if the page looks smoother.

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- Can the result be reproduced from saved parameters and source data?

Archival handling and metadata

The original page should remain immutable. Store the original file, enhanced grayscale image, binary text mask, page crop, defect overlay and processing metadata. Metadata should include source filename, capture date if known, image dimensions, software version, parameters, elapsed time and any manual correction. This supports reproducibility and audit.

Practical design implications

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Colour handling for blue annotations

Blue notes can be weak in grayscale because conversion combines colour channels. Compare grayscale, blue channel, red channel, HSV saturation and Lab colour components on actual images. A channel should be selected based on annotation contrast and noise. Keep a colour crop even if OCR uses a grayscale or binary derivative.

Practical design implications

The implementation should use a conservative, reversible approach. Every enhancement stage should produce a separate output so that staff can compare it with the original. If a method improves one metric while damaging another, record the trade-off and select the stage based on the primary requirement of readable, faithful document preservation.

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Perspective and page geometry

A page photographed obliquely is a quadrilateral rather than a rectangle. Detect long edges, estimate corners in consistent top-left, top-right, bottom-right, bottom-left order, compute a homography and warp the image. If a corner is hidden, use line intersections or a constrained quadrilateral fit, and flag uncertain cases for manual review.

Practical design implications

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Runtime optimization

Measure stage-level latency using a monotonic high-resolution timer. Avoid repeated colour conversions, unnecessary copies and Python loops over every pixel. Prefer vectorized NumPy and OpenCV functions. Resize early for detection, but apply the geometric transform carefully to the source or a sufficiently detailed working image.

Practical design implications

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- Can the result be reproduced from saved parameters and source data?

Quality assurance and acceptance criteria

A page should pass if text Dice is at least 0.90, page corners are within a chosen tolerance, OCR CER does not worsen materially, required annotations are retained and processing time meets the task limit. Pages with low confidence should be flagged for review rather than silently output as successful.

Practical design implications

The implementation should use a conservative, reversible approach. Every enhancement stage should produce a separate output so that staff can compare it with the original. If a method improves one metric while damaging another, record the trade-off and select the stage based on the primary requirement of readable, faithful document preservation.

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Ethical and technical risks of learned restoration

A learned model can create plausible details that were not present in the source, especially in faint or damaged text. This is unacceptable for archival fidelity unless the original is preserved and the output is clearly labelled as reconstructed. Compare models using identical pages, report inference hardware and check for hallucinated strokes.

Practical design implications

The implementation should use a conservative, reversible approach. Every enhancement stage should produce a separate output so that staff can compare it with the original. If a method improves one metric while damaging another, record the trade-off and select the stage based on the primary requirement of readable, faithful document preservation.

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